Ex 5.5.2

Suppose that D is a domain and $f \in H(D)$ is nonvanishing as well as nonconstant on D. Then $z \mapsto |f(z)|$ has no local minimums on D.

Pf. Unlike Example 5.41, we have f is nonvanishing. We exploit f's nonvanishing property to pass from a minimum of $|f|$ to a maximum of $|\frac{1}{f}|$.

Using Prop 2.2 (Quotient Rule) the function, $g(z) := \frac{1}{f(z)}$, for $z \in D$, is holomorphic on D. Also since f is nonconstant, g is also nonconstant.

Assume, for proof by contradiction, that $z \mapsto |f(z)|$ has a local minimum at $z_0 \in D$. Then $\exists \varepsilon > 0$ with $D_\varepsilon(z_0) \subseteq D$ and $|f(z)| \geqslant |f(z_0)| > 0$, $\forall z \in D_\varepsilon(z_0)$. The $\forall z \in D_\epsilon(z_0)$:

$$|g(z)| = \frac{1}{|f(z)|} \leqslant \frac{1}{|f(z_0)|} = |g(z_0)|$$

therefore $z \mapsto |g(z)|$ has a local maximum at z_0. But this contradicts Thm 5.40 (Maximum Modulus Theorem) applied to the nonconstant $g \in H(D)$.

Therefore, $z \mapsto |f(z)|$ has no local minimum on D. □

Ex 5.5.3

Let (f_n) be a sequence of nonvanishing functions in $H(D)$ for a domain D. Suppose that $f_n \to f$ uniformly on compact subsets of D. Show that either f is identically zero or f never vanishes on D. (This is known as Hurwitz's Theorem.)

Pf. We are given that each $f_n \in H(D)$ and $f_n \to f$ uniformly. Using Thm 5.32, we have also that $f \in H(D)$ and $f_n' \to f'$ on compact subsets of D.
Assume f is not identically zero. We show f never vanishes on D.

Assume, for proof by contradiction, $\exists a \in D$ s.t. $f(a) = 0$. Since f is not identically zero, f is nonconstant, using Thm 5.6 we have the zero set $Z(f)$ is discrete in D and a is a zero of finite order.
Take $r > 0$ with $\overline{D_r(a)} \subseteq D$ and $f(z) \neq 0$ for $z \in \overline{D_r(a)} \setminus \{a\}$.

Let $\gamma(t) = a + re^{it}$ for $t \in [0, 2\pi]$. Then γ^* is compact and disjoint from $Z(f)$, thus,

$$m := \min_{z \in \gamma^*} |f(z)| > 0.$$

By uniform convergence on the compact set γ^*, $\exists N \in \mathbb{N}$ s.t. $|f_n(z) - f(z)| < m/2$, $\forall z \in \gamma^*$ and $n \geqslant N$. Using the reverse triangle inequality for $z \in \gamma^*$ and $n \geqslant N$, we have,

$$|f_n(z)| \geqslant |f(z)| - |f_n(z) - f(z)| \geqslant m - m/2 = m/2 > 0$$

Therefore, both $|f|$ and $|f_n|$ are bounded below by $m/2$ for $n \geqslant N$ on γ^*. Thus, the convergences $f_n \to f$ and $f_n' \to f'$ are uniform on γ^*. Using a standard estimate, we have,

$$\frac{f_n'}{f_n} \to \frac{f'}{f}$$

uniformly on γ^*.

Take any $R > r$ and $\overline{D_R(a)} \subseteq D$. Using Thm 5.35 in the convex domain $D_R(a)$, we have: $\forall h \in H(D_R(a))$ with finitely many zeros of which none are on γ^*,

$$\frac{1}{2\pi i} \int_\gamma \frac{h'(w)}{h(w)}\, dw = N_h$$

where N_h is the number of zeros of h inside γ, counted with multiplicity (since $\mathrm{Ind}_\gamma = 1$ on its interior).
We have $\forall n \geqslant N$, f_n is nonvanishing on D. Therefore, $N_{f_n} = 0$, and

$$\frac{1}{2\pi i} \int_\gamma \frac{f_n'(w)}{f_n(w)}\, dw = 0$$

By uniform convergence of $\dfrac{f_n'}{f_n} \to \dfrac{f'}{f}$ on γ^*, we have the limit:

$$N_f = \frac{1}{2\pi i} \int_\gamma \frac{f'(w)}{f(w)}\, dw = \lim_{n \to \infty} \frac{1}{2\pi i} \int_\gamma \frac{f_n'(w)}{f_n(w)}\, dw = 0$$

But since a is a zero of f inside γ of order $\geqslant 1$, we must have $N_f \geqslant 1$, which gives us a contradiction.

Therefore, f never vanishes on D. □

Ex 5.5.4

Suppose that D is a bounded domain and that f is continuous on the closure $\overline{D}$ and holomorphic and nonconstant on D. If $|f(z)|$ is constant on the boundary ∂D, then show that f has a zero in D.

Pf. Let $|f(z)| = c$, $\forall z \in \partial D$, with $c \geqslant 0$. Then we can consider:

- Case $c = 0$: Then $f \equiv 0$ on ∂D. By Cor 5.42 applied to the bounded domain D with f continuous on $\overline{D}$ and holomorphic on D, we have that f attains its maximum on the boundary ∂D of D, i.e.

  $$\max_{z \in \overline{D}} |f(z)| = \max_{z \in \partial D} |f(z)| = 0,$$

  so $f \equiv 0$ on $\overline{D}$. This contradicts our assumption of f being nonconstant on D.

- Case $c > 0$: Assume, for proof by contradiction, that f has no zeros in D. Using the fact $|f| = c > 0$ on ∂D, this gives $f \neq 0$ on all of $\overline{D}$. Therefore, using Prop 2.2 again, we have $g := \dfrac{1}{f}$ is continuous on $\overline{D}$, holomorphic, and nonconstant on D.

  Now apply Cor 5.42 to both f and g:

  $$\max_{z \in \overline{D}} |f(z)| = \max_{z \in \partial D} |f(z)| = c$$
  $$\max_{z \in \overline{D}} |g(z)| = \max_{z \in \partial D} \frac{1}{|f(z)|} = \frac{1}{c}$$

  This gives that $|f(z)| \leqslant c$ on $\overline{D}$, and simultaneously $|f(z)| \geqslant c$ on $\overline{D}$. Therefore, $|f| \equiv c$ on $\overline{D}$, in particular on the domain D.

  By Ex 2.2.4, since $f \in H(D)$ and $z \mapsto |f(z)|$ is constant on D, f must be constant on D. This contradicts our assumption.

Therefore f has a zero in D. □

Ex 5.5.6

Suppose that f is entire and one-to-one. Show that $f(z) = az + b$ with $a \neq 0$.

Pf. Let $g : \mathbb{C} \setminus \{0\} \to \mathbb{C}$ s.t.

$$g(z) := f(\frac{1}{z})$$

Since f is entire and $z \mapsto 1/z$ is holomorphic on $\mathbb{C} \setminus \{0\}$, by Prop 2.2 (Chain Rule) g is holomorphic on $\mathbb{C} \setminus \{0\}$, so 0 is an isolated singularity of g. Using Def 5.20, we can consider the singulrity types for g:

- Removable: If 0 is removable for g, then by Prop 5.13(d) $\exists r, M > 0$ s.t. $|g(z)| \leqslant M$ for $z \in D'_r(0)$, i.e. $|f(w)| \leqslant M$ for $|w| > 1/r$. Combined with the continuity of f on $\overline{D_{1/r}(0)}$, f is bounded on $\mathbb{C}$.

  Then, using Thm 5.26 (Liouville), f is constant, which contradicts injectivity.

- Essential: If 0 is essential for g, then by Thm 5.21 (Casorati-Weierstrass), $\forall \rho > 0$ the image $g(D'_\rho(0))$ is dense in $\mathbb{C}$. I.e., $f(\{|w| > 1/\rho\})$ is dense in $\mathbb{C}$ for every $\rho > 0$.

  Take $\rho = 1$, s.t. $f(\{|w| > 1\})$ is dense. But Thm 5.38 (Open Mapping Theorem) applied to the nonconstant entire f on the domain $D_1(0)$ gives, the set $f(D_1(0))$ is open and nonempty (contains $f(0)$). A dense subset of $\mathbb{C}$ meets every nonempty open set, $\exists w_1 \in \{|w| > 1\}$ and $\exists w_2 \in D_1(0)$ with $f(w_1) = f(w_2)$. Since $|w_1| > 1 > |w_2|$, $w_1 \neq w_2$, which also contradicts injectivity.

- Pole: The remaining case gives that g must have a pole, of order $m \geqslant 1$, at 0. This means f has a pole of order m at ∞, so by Lemma 5.23, f is a polynomial of degree m.

Assume, $m \geqslant 2$. Then, the polynomial $f(z) - f(0)$ has degree m, hence by the Fundamental Theorem of Algebra (Ex 5.3.7) it has exactly $m \geqslant 2$ roots in $\mathbb{C}$, counted with multiplicity.
If two of these roots are distinct, f takes the value $f(0)$ at two distinct points, which contradicts injectivity again. Otherwise all m roots coincide at $z_0 \in \mathbb{C}$, which gives z_0 is a zero of order $m \geqslant 2$ of $f - f(0)$.

By Thm 5.37 (Local Behavior), $\exists \varepsilon > 0$ s.t. w sufficiently close (but not equal) to 0, the equation $f(z) - f(0) = w$ has $m \geqslant 2$ distinct solutions in $D_\varepsilon(z_0)$, which again contradicts injectivity.

Therefore $m = 1$, so $f(z) = az + b$ with $a \neq 0$. □

LH-1

Recall from calculus that the Mean Value Theorem says that if f is a continuous real-valued function on $[a, b]$ and if f is differentiable on (a, b), then there is a $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

(a) Show that the Mean Value Theorem need not hold for holomorphic functions—consider $f(z) = \exp(z)$ and $[a, b] = [0, 2\pi i]$.

(b) Nevertheless, show that if $|f'(z)| \leqslant M$ for all $z \in [w_1, w_2]$, then

$$|f(w_2) - f(w_1)| \leqslant M|w_2 - w_1|.$$

Pf.

(a) Let $f(z) = \exp(z)$. Using Prop 2.16, f is entire and $f'(z) = \exp(z)$. Therefore, we have

$$\frac{f(2\pi i) - f(0)}{2\pi i - 0} = \frac{e^{2\pi i} - e^0}{2\pi i} = \frac{1 - 1}{2\pi i} = 0$$

i.e. any c satisfying the Mean Value Theorem identity needs $f'(c)(2\pi i - 0) = 0 \implies \exp(c) = 0$. But $|\exp(c)| = e^{\mathrm{Re}(c)} > 0$, $\forall c \in \mathbb{C}$. Thus, no such c exists and the Mean Value Theorem fails for f on $[0, 2\pi i]$.

(b) Take $|f'(z)| \leqslant M$ on the line segment $[w_1, w_2]$, in particular f' is defined (and holomorphic, by Cor 4.13, hence continuous) on this segment.

We can parameterize the segment by $\gamma : [0, 1] \to \mathbb{C}$ s.t. $\gamma(t) = w_1 + t(w_2 - w_1)$. Then $\gamma'(t) = w_2 - w_1$ and, by Prop 2.2 (Chain Rule), the composition $t \mapsto f(\gamma(t))$ is differentiable on $[0, 1]$ with derivative $f'(\gamma(t))(w_2 - w_1)$.

Using the Fundamental Theorem of Calculus applied to the real and imaginary parts,

$$f(w_2) - f(w_1) = \int_0^1 f'(\gamma(t))(w_2 - w_1)\, dt = \int_\gamma f'(w)\, dw$$

Now we apply the standard path-integral estimate from Prop 3.26 to obtain the required form,

$$|f(w_2) - f(w_1)| = \left| \int_\gamma f'(w)\, dw \right| \leqslant \max_{z \in [w_1, w_2]} |f'(z)| \mathrm{length}(\gamma) \leqslant M|w_2 - w_1|$$

LH-2

(The Indeterminate Form $\frac{0}{0}$—Baby Version.) Suppose that D is a domain, $a \in D$, and that f and g are nonconstant functions in $H(D)$ such that $f(a) = 0 = g(a)$ and $g'(a) \neq 0$. Show that

$$\lim_{z \to a} \frac{f(z)}{g(z)} = \frac{f'(a)}{g'(a)} = \lim_{z \to a} \frac{f'(z)}{g'(z)}.$$

Hint: $\dfrac{f(z)}{g(z)} = \dfrac{(f(z) - f(a))/(z-a)}{(g(z) - g(a))/(z-a)}$.

Pf. Given $g'(a) \neq 0$ and g is continuous (using Cor 4.13 as above), $\exists r > 0$ s.t. $g(z) \neq g(a) = 0$, $\forall z \in D'_r(a)$. This is true since otherwise the difference quotient for $g'(a)$ would have a sequence with limit 0, giving $g'(a) = 0$). On $D'_r(a)$ we can divide and use $f(a) = g(a) = 0$:

$$\frac{f(z)}{g(z)} = \frac{f(z) - f(a)}{g(z) - g(a)} = \frac{\dfrac{f(z) - f(a)}{z - a}}{\dfrac{g(z) - g(a)}{z - a}}.$$

By Def 2.1, as $z \to a$ the numerator tends to $f'(a)$ and the denominator tends to $g'(a) \neq 0$. By Prop 2.2 (Quotient Rule),

$$\lim_{z \to a} \frac{f(z)}{g(z)} = \frac{f'(a)}{g'(a)}$$

Now, using Cor 4.13, both $f', g' \in H(D)$, and in particular both are continuous on D. Since $g'(a) \neq 0$, $\exists r' > 0$ s.t. $g'(z) \neq 0$ on $D_{r'}(a)$. By Prop 2.2, on this disk we have $\dfrac{f'}{g'}$ is continuous. Therefore,

$$\lim_{z \to a} \frac{f'(z)}{g'(z)} = \frac{f'(a)}{g'(a)}$$

LH-3

(The Indeterminate Form $\frac{0}{0}$—Adult Version.) Suppose that D is a domain, $a \in D$, and that f and g are nonconstant functions in $H(D)$ such that $f(a) = 0 = g(a)$. We want to investigate

$$\lim_{z \to a} \frac{f(z)}{g(z)}.$$

Note that f and g must have zeros of finite order m and n, respectively, at a.

(a) What kind of isolated singularity does $h = f/g$ have at a? (Your answer will depend on m and n.)

(b) If $m = n$, show that

$$\lim_{z \to a} \frac{f(z)}{g(z)} = \frac{f^{(m)}(a)}{g^{(m)}(a)}.$$

Pf. By Thm 5.6, since f and g are given as nonconstant, their zeros at a have finite order, $m \geqslant 1$ and $n \geqslant 1$. By Lemm 5.4, $\exists F, G \in H(D)$ with $F(a) \neq 0$ and $G(a) \neq 0$, s.t.

$$f(z) = (z-a)^m F(z)$$
$$g(z) = (z-a)^n G(z)$$

$\forall z \in D$. Also, the proof for Lemm 5.4 gives,

$$F(a) = \frac{f^{(m)}(a)}{m!}$$
$$G(a) = \frac{g^{(n)}(a)}{n!}$$

By continuity, since $G(a) \neq 0$, $\exists r > 0$ with $\overline{D_r(a)} \subseteq D$ and $G(z) \neq 0$ on $D_r(a)$. Then $g(z) = 0$ in $D_r(a)$ at $z = a$, By Prop 2.2, $h = f/g$ is well-defined and holomorphic on $D_r(a) \setminus \{a\}$. Hence it has an isolated singularity at a. On $D_r(a) \setminus \{a\}$, we have,

$$h(z) = \frac{(z-a)^m F(z)}{(z-a)^n G(z)} = (z-a)^{m-n} \varphi(z)$$

where $\varphi(z) := F(z)/G(z) \in H(D_r(a))$ and $\varphi(a) = F(a)/G(a) \neq 0$ using Prop 2.2.

(a) We consider the singularity of h at a when:

- $\underline{m > n}$: $h(z) = (z-a)^{m-n}\varphi(z)$, and the RHS is holomorphic on $D_r(a)$, with 0 at a.
  Therefore, a is a removable singularity of h. Applying Lemm 5.4 to this holomorphic extension, the extended h has a zero of order $m - n$ at a.
- $\underline{m = n}$: $h(z) = \varphi(z)$ on $D_r(a) \setminus \{a\}$, which gives a holomorphic extension to $D_r(a)$ with value $\varphi(a) = F(a)/G(a) \neq 0$.
  Therefore, a is a removable singularity of h, with a nonzero limit.
- $\underline{m < n}$: $h(z) = \varphi(z)/(z-a)^{n-m}$ with $\varphi \in H(D_r(a))$ and $\varphi(a) \neq 0$. Using Cor 5.18, a is a pole of h of order $n - m$.

(b) When $m = n$, using the case for $m = n$ in part (a), the limit at a exists and can be expressed as,

$$\lim_{z \to a} \frac{f(z)}{g(z)} = \lim_{z \to a} \varphi(z) = \frac{F(a)}{G(a)} = \frac{f^{(m)}(a)/m!}{g^{(m)}(a)/m!} = \frac{f^{(m)}(a)}{g^{(m)}(a)}$$

LH-4

(The Indeterminate Form $\frac{\infty}{\infty}$.) Suppose that D is a domain, $a \in D$, and that f and g are functions in $H(D \setminus \{a\})$ such that f and g have poles of order m and n, respectively, at a. We want to investigate

$$\lim_{z \to a} \frac{f(z)}{g(z)}.$$

(a) What kind of isolated singularity does $h = f/g$ have at a? (Your answer will depend on m and n.)

(b) Suppose that $m = n$ and that $\sum_{j=1}^{n} \frac{b_j}{(z-a)^j}$ is the principal part of f at a and that $\sum_{j=1}^{n} \frac{c_j}{(z-a)^j}$ is the principal part of g at a. Show that

$$\lim_{z \to a} \frac{f(z)}{g(z)} = \frac{b_n}{c_n}.$$

Pf. By Cor 5.18, since f has a pole of order m at a and g has a pole of order n at a, $\exists F, G \in H(D)$ with $F(a) \neq 0$ and $G(a) \neq 0$ s.t.

$$f(z) = \frac{F(z)}{(z-a)^m}$$
$$g(z) = \frac{G(z)}{(z-a)^n}$$

$\forall z \in D \setminus \{a\}$. By continuity, take $r > 0$ with $\overline{D_r(a)} \subseteq D$, and $G(z) \neq 0$ on $D_r(a)$. Then $\forall z \in D_r(a) \setminus \{a\}$, $g(z) \neq 0$. Therefore, by Prop 2.2, $h = f/g$ is holomorphic and has an isolated singularity at a. We have,

$$h(z) = \frac{F(z)/(z-a)^m}{G(z)/(z-a)^n} = (z-a)^{n-m}\varphi(z)$$

where $\varphi(z) := F(z)/G(z) \in H(D_r(a))$ and $\varphi(a) = F(a)/G(a) \neq 0$, as in LH-3.

(a) We consider the singularity of h at a when:

- $\underline{m > n}$: $h(z) = \varphi(z)/(z-a)^{m-n}$ with $\varphi(a) \neq 0$. Using Cor 5.18, a is a pole of h of order $m - n$.
- $\underline{m = n}$: $h(z) = \varphi(z)$ on $D_r(a) \setminus \{a\}$, which has a holomorphic extension to $D_r(a)$ with nonzero value $\varphi(a)$.
  Therefore, a is a removable singularity of h with a nonzero limit.
- $\underline{m < n}$: $h(z) = (z-a)^{n-m}\varphi(z)$ has a holomorphic extension to $D_r(a)$ with value 0 at a.
  Therefore, a is a removable singularity. Applying Lemm 5.4 to this holomorphic extension, the extended h has a zero of order $n - m$ at a.

(b) When $m = n$, use Def 5.14 (since $\sum_{j=1}^{n} b_j/(z-a)^j$ is the principal part of f at a) to write,

$$r_f(z) := f(z) - \sum_{j=1}^{n} \frac{b_j}{(z-a)^j}$$

with a removable singularity at a with $b_n \neq 0$. Multiplying by $(z-a)^n$,

$$(z-a)^n f(z) = \sum_{j=1}^{n} b_j (z-a)^{n-j} + (z-a)^n r_f(z)$$

Using the holomorphic extension of r_f, the RHS extends to a holomorphic function on D, $F(z)$. Then we have $F \in H(D)$, therefore $f(z) = F(z)/(z-a)^n$ on $D \setminus \{a\}$. Evaluating at $z = a$ gives $F(a) = b_n$.

The same argument gives $g(z) = G(z)/(z-a)^n$ with $G \in H(D)$ and $G(a) = c_n$.

By the uniqueness of representation in Cor 5.18 (i.e., F and G are exactly the holomorphic functions at the start of this solution), we have on $D_r(a) \setminus \{a\}$ as $z \to a$,

$$\frac{f(z)}{g(z)} = \frac{F(z)}{G(z)} \to \frac{F(a)}{G(a)} = \frac{b_n}{c_n}$$

using continuity of F/G at a by Prop 2.2, given $G(a) = c_n \neq 0$. □

EP-3

Verify the claim made in lecture during the proof of the general Cauchy Theorem: suppose D is a domain and that g is continuous on $D \times D$. If $z_n \to z$ in D and if $K \subseteq D$ is compact, show that $f_n(\cdot) = g(z_n, \cdot)$ converges uniformly on K to $f(\cdot) = g(z, \cdot)$. (Hint: there are lots of ways to approach this problem. I used the fact that a continuous function on a compact metric space is necessarily uniformly continuous—a reference is Theorem 4.19 in Rudin's *Principles of Mathematical Analysis* or Theorem 1.33 of Folland's *Advanced Calculus*.)

Pf. Take a compact subset of $D \times D$ where g is uniformly continuous, and then use that to extract the uniform estimate on K.
Let

$$A := \{z_n : n \in \mathbb{N}\} \cup \{z\} \subseteq D.$$

We can show that A is compact. It is bounded since the convergent sequence (z_n) is bounded, and closed since any limit point of A is the limit of a subsequence of (z_n) (or a constant subsequence equal to z), and every subsequence of (z_n) converges to $z \in A$. Using the Heine-Borel Theorem, we have A is compact. Also, $A \subseteq D$.

Then $A \times K \subseteq D \times D$ is compact, as the product of two compact sets in $\mathbb{C}$ (which works for $\mathbb{R}^2$ also).
Now we add the product metric, $d\big((z_1, w_1), (z_2, w_2)\big) := \max(|z_1 - z_2|, |w_1 - w_2|)$, or equivalent to $\mathbb{C} \times \mathbb{C}$. The restriction $g|_{A \times K}$ is continuous on the compact metric space $A \times K$, is hence uniformly continuous (by Rudin's Theorem 4.19).

Let $\varepsilon > 0$, and choose $\delta > 0$ s.t. $\forall (\zeta_1, w_1), (\zeta_2, w_2) \in A \times K$, we have,

$$d\big((\zeta_1, w_1), (\zeta_2, w_2)\big) < \delta \implies |g(\zeta_1, w_1) - g(\zeta_2, w_2)| < \varepsilon$$

Since $z_n \to z$, there exists $N \in \mathbb{N}$ such that $|z_n - z| < \delta$ for all $n \geqslant N$. Then $\forall n \geqslant N$ and $\forall w \in K$, the points (z_n, w) and (z, w) both lie in $A \times K$, and,

$$\begin{aligned} &d\big((z_n, w), (z, w)\big) = |z_n - z| < \delta \\ \implies &|f_n(w) - f(w)| = |g(z_n, w) - g(z, w)| < \varepsilon \end{aligned}$$

This bound is independent of $w \in K$. Therefore, $f_n \to f$ uniformly on K. □